A TWO-INVOLUTION CONSTRUCTION OF WING BIJECTIONS

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Abstract. For a finite subset of the Boolean cube with an assigned wing at each point, we construct a permutation that changes every assigned wing coordinate and whose agreement count in each coordinate is bounded by the corresponding boundary count. The construction alternates two explicit involutions on a finite auxiliary state space. A normalization involution with a unique fixed point and a coordinatewise weight invariant establish termination and correctness. The argument uses no determinants or linear algebra and yields Reimer's Butterfly Lemma on the whole cube as a corollary. It gives a combinatorial construction for Turturean's wing-bijection theorem, proved in his work on the Aaronson–Ambainis conjecture. Both results are formalized in Lean 4.

1. Introduction

Turturean [8, Theorem 3.1] proved that a finite set of cube points with prescribed wings admits a bijection satisfying both the wing constraints and coordinatewise boundary bounds. His proof extracts a permutation from a non-vanishing determinant. Section 9.5 of that paper asks for a direct combinatorial construction. Theorem 2.1 gives such a construction.

We use the involution principle of Garsia and Milne [3]. Normalization of interior cube edges supplies a unique starting state for an alternating walk. A coordinatewise weight invariant bounds the state space and gives the output conditions (Lemma 5.3). Property (a) of Theorem 2.1 also yields Reimer's Butterfly Lemma (Section 7), whose proofs by Reimer [6] and Borgs, Chayes and Randall [2] use linear algebra.

In Turturean's proof [8] of the Aaronson–Ambainis conjecture [1], the determinant is used only to establish the wing-bijection theorem. Subsequent steps use its counting consequence, Turturean's Corollary 3.2. Substituting our construction therefore removes the determinant argument from that proof.

The wing-bijection theorem and the Butterfly Lemma corollary are formalized in Lean 4 using mathlib [4, 5]. The source and an executable implementation of the construction are available at `https://github.com/hxrdxkxvxd/wing-bijection`.

2. Setting and construction theorem

We write points of the cube as $x \in \{0,1\}^n$ with coordinates indexed by $[n] = \{0, 1, \dots, n-1\}$. For $S \subseteq [n]$, let x^S be x with the coordinates in S changed, and write $x^j = x^{\{j\}}$. Let $\Delta(x, y) = \{i : x_i \neq y_i\}$.

Fix $n \geq 0$ and a set $\mathcal{O} \subseteq \{0,1\}^n$ with $N = |\mathcal{O}|$.

- A *wing* at $x \in \mathcal{O}$ is a set $A \subseteq [n]$ such that $x^S \in \mathcal{O}$ for every $S \subseteq A$. A wing assignment is a choice of a wing A_x at every $x \in \mathcal{O}$. The empty set is always a wing.
- An *interior j-edge* is a pair $\{x, x^j\} \subseteq \mathcal{O}$. Let E_j be the set of interior j-edges and $e_j = |E_j|$. A point $x \in \mathcal{O}$ is *interior* in direction j if $x^j \in \mathcal{O}$; otherwise it is a *j-boundary point*.
- Let b_j be the number of j-boundary points. Then $N = 2e_j + b_j$.
- For a map $p\colon \mathcal{O} \to \mathcal{O}$ put

$$a_j(p) = \#\{x \in \mathcal{O} :\ p(x)_j = x_j\}, \qquad d_j(p) = \#\{x \in \mathcal{O} :\ p(x)_j \neq x_j\} = N - a_j(p).$$

Throughout, $[\,P\,]$ denotes 1 if P holds and 0 otherwise. We compose permutations as maps: $(\rho \circ \gamma)(x) = \rho(\gamma(x))$.

Date: November 2026.

Theorem 2.1 (Wing bijection [8, Theorem 3.1], constructive form)**.** *Let $\mathcal{O} \subseteq \{0,1\}^n$ and let $(A_x)_{x\in\mathcal{O}}$ be a wing assignment. Then there is a permutation τ of $\mathcal{O}$ such that*

(a) *$\tau(x)_i \neq x_i$ for every $x \in \mathcal{O}$ and every $i \in A_x$, and*

(b) *$a_j(\tau) \leq b_j$ for every $j \in [n]$.*

Moreover, the permutation τ is produced by the deterministic procedure of Section 6.

Empty sums and empty compositions have their usual values, so the construction also covers $n = 0$ and $\mathcal{O} = \varnothing$.

All choices below use fixed linear orders, chosen in advance, on coordinates, points, pairs of points, and edges. "The first" element always refers to these orders, so every map we define is deterministic.

3. Alternating two involutions

Lemma 3.1. *Let X be a finite set and let $L, R\colon X \to X$ be involutions. Suppose that L has exactly one fixed point c. Define $c_0 = c$ and $c_{k+1} = L(R(c_k))$. Then $R(c_k) = c_k$ for some $k \geq 0$.*

Proof. Form a multigraph G on X with one L-edge $\{y, L(y)\}$ for each two-element orbit of L and one R-edge $\{y, R(y)\}$ for each two-element orbit of R. A pair that is an orbit of both involutions gives two parallel edges of different colours. Every vertex meets at most one edge of each colour, so its degree is at most 2. The vertex c meets no L-edge, so its degree is at most 1.

A finite connected graph of maximum degree 2 that contains a vertex of degree at most 1 is a path. Hence the component of c is a path with endpoint c. Its edges alternate in colour, and the edge at c, if there is one, is an R-edge. The sequence $c_0,\ R(c_0),\ c_1,\ R(c_1), \dots$ therefore walks along this path. If the path is the single vertex c, then $R(c) = c$ and we are done. Otherwise the walk reaches the other endpoint $w \neq c$ after finitely many steps. The vertex w has degree 1, so it is fixed by L or by R. It is not fixed by L, because $w \neq c$. Hence $R(w) = w$.

Since w has no R-edge, the walk arrives at w through an L-edge, so $w = c_k$ for some k. □

We also use a standard way of extending an involution along its fixed points.

Lemma 3.2. *Let T be an involution of a set X and let $f\colon \mathrm{Fix}(T) \to Y$ be a bijection. Let J_Y be an involution of Y with exactly one fixed point y_0. Define $J\colon X \to X$ by $J(x) = T(x)$ if $T(x) \neq x$, and $J(x) = f^{-1}(J_Y(f(x)))$ if $T(x) = x$. Then J is an involution of X whose only fixed point is $f^{-1}(y_0)$.*

Proof. The map T sends non-fixed points to non-fixed points, and $f^{-1} \circ J_Y \circ f$ is an involution of $\mathrm{Fix}(T)$. Therefore J is an involution. A non-fixed point of T is not fixed by J. A fixed point x of T is fixed by J exactly when $f(x) = y_0$. □

4. The normalization involution

Let σ_j be the permutation of $\mathcal{O}$ that exchanges x and x^j for every interior j-edge $\{x, x^j\}$ and fixes every j-boundary point. It is an involution, it changes only coordinate j, and

$$\sigma_j(x)_j \neq x_j \iff x \text{ is interior in direction } j. \tag{1}$$

We write each edge of E_j as $\{u, v\}$ with $u_j = 0$ and $v_j = 1$.

Definition 4.1. For $0 \leq i \leq n$, let C_i be the set of pairs (ρ, k) such that

(C1) $\rho \in \mathrm{Sym}(\mathcal{O})$ and $\rho(x)_j = x_j$ for all $x \in \mathcal{O}$ and all $j < i$;

(C2) k assigns a nonnegative integer k_e to each edge $e \in E_j$ with $j \geq i$;

(C3) $a_j(\rho) + 2\sum_{e\in E_j} k_e = b_j$ for all $j \geq i$.

The set C_n has exactly one element, $(\mathrm{id}, \varnothing)$. A permutation that preserves every coordinate is the identity, and for $i = n$ there are no counters and no constraints.

The level involutions. Let $i < n$ and $(\rho, k) \in C_i$. Call an edge $e = \{u, v\} \in E_i$ *normal* if

$$\rho(u)_i = 1, \qquad \rho(v)_i = 0, \qquad k_e = 0.$$

If every edge of E_i is normal, let $T_i(\rho, k) = (\rho, k)$. Otherwise let $e = \{u, v\}$ be the first edge that is not normal. Let $T_i(\rho, k) = (\rho', k')$, where $\rho' = \rho \circ (u\, v)$, that is, $\rho'(u) = \rho(v)$ and $\rho'(v) = \rho(u)$. The counters are unchanged except for k'_e, which is given by Table 1.

$(\rho(u)_i,\ \rho(v)_i)$	new counter k'_e
$(0, 1)$	$k_e + 1$
$(1, 0)$	$k_e - 1$
$(0, 0)$ or $(1, 1)$	k_e

TABLE 1. Counter update of T_i on the first non-normal edge $e = \{u, v\}$. In the case $(1, 0)$, $k_e > 0$ because e is not normal.

Lemma 4.2. *T_i is an involution of C_i. Its fixed points are the pairs in which every edge of E_i is normal.*

Proof. T_i maps C_i to itself. Condition (C1) holds because u and v agree in every coordinate $j < i$, so exchanging their images preserves $\rho(x)_j = x_j$. For $j \neq i$, the points u and v also agree in coordinate j, so $a_j(\rho') = a_j(\rho)$ and (C3) is untouched.

For $j = i$, compare the bits before and after the move. In case $(0, 1)$ both u and v keep coordinate i before the move and both change it after, so a_i drops by 2 while $2\sum_{E_i} k$ rises by 2. Case $(1, 0)$ is the reverse; there $k_e > 0$, since otherwise e would be normal. In the remaining cases a_i does not change.

T_i is an involution. A move changes only the images of u and v and the counter k_e, so the normality of every other edge of E_i is unchanged. The edges of E_i are pairwise disjoint. The edge e is still not normal after the move: the move turns the bits $(0, 1)$ into $(1, 0)$ with $k'_e > 0$, turns $(1, 0)$ into $(0, 1)$, and leaves $(0, 0)$ and $(1, 1)$ alone. Hence the second application acts on the same edge, and by Table 1 it restores ρ and k_e. A move changes ρ, so the fixed points are exactly the pairs in which every edge is normal. □

Lemma 4.3. *The map $f_i(\rho, k) = (\rho \circ \sigma_i,\ k|_{\bigcup_{j>i} E_j})$ is a bijection from $\mathrm{Fix}(T_i)$ onto C_{i+1}. Its inverse is $(\rho', k') \mapsto (\rho' \circ \sigma_i,\ k' \cup 0_{E_i})$, where 0_{E_i} puts zero counters on E_i.*

Proof. Let (ρ, k) be fixed by T_i. All counters on E_i vanish, and every interior point in direction i changes coordinate i under ρ. By (C3) we have $a_i(\rho) = b_i$, so every i-boundary point keeps coordinate i.

By (1), $\rho \circ \sigma_i$ therefore keeps coordinate i at every point. It also keeps every coordinate $j < i$, because ρ and σ_i do. For $j > i$, the permutation σ_i preserves coordinate j. Substituting $y = \sigma_i(x)$ gives $a_j(\rho \circ \sigma_i) = a_j(\rho)$, so (C3) still holds. Thus f_i maps into C_{i+1}.

Conversely, let $(\rho', k') \in C_{i+1}$. Since ρ' preserves coordinate i, (1) shows that $\rho' \circ \sigma_i$ changes coordinate i exactly at the $2e_i$ interior points. Hence $a_i(\rho' \circ \sigma_i) = N - 2e_i = b_i$, and (C3) holds for $j = i$ with zero counters. Every edge $\{u, v\} \in E_i$ is normal, since $(\rho'\sigma_i)(u)_i = v_i = 1$ and $(\rho'\sigma_i)(v)_i = u_i = 0$. The remaining conditions are checked as before. The two maps are mutually inverse because $\sigma_i \circ \sigma_i = \mathrm{id}$. □

Let J_n be the identity of C_n. For $i = n - 1, \ldots, 0$, let J_i be the involution of C_i obtained from T_i, f_i and J_{i+1} by Lemma 3.2. Put $J = J_0$. Explicitly, J applies $T_0, T_1, \ldots$ in turn, passing to the next level through f_i whenever the current T_i fixes the pair. It performs the first non-trivial move it finds and maps the result back to C_0.

Proposition 4.4. *J is an involution of C_0 whose only fixed point is $(\pi_*, 0)$, where $\pi_* = \sigma_{n-1} \circ \cdots \circ \sigma_0$. In particular $a_j(\pi_*) = b_j$ for every j.*

Proof. By downward induction using Lemma 3.2, each J_i is an involution of C_i with a unique fixed point. The fixed point of J_n is $(\mathrm{id}, \varnothing)$. Pulling it back through $f_{n-1}^{-1}, \ldots, f_0^{-1}$ multiplies on the right by $\sigma_{n-1}, \ldots, \sigma_0$ and adds zero counters. Condition (C3) for C_0 with zero counters gives $a_j(\pi_*) = b_j$. □

5. States and the two involutions

Call a permutation γ of $\mathcal{O}$ *local* if $\Delta(x, \gamma(x)) \subseteq A_x$ for every x.

Definition 5.1. A *state* is a tuple $c = (S, \pi, \Gamma, k)$ consisting of

(i) subsets $S_x \subseteq A_x$ for $x \in \mathcal{O}$;
(ii) a permutation π of $\mathcal{O}$;
(iii) a finite sequence $\Gamma = (\gamma_1, \ldots, \gamma_r)$ of local permutations of $\mathcal{O}$ different from id (the *stack*);
(iv) a nonnegative integer k_e for every interior edge $e \in \bigcup_j E_j$.

Put $\alpha(x) = x^{S_x}$. Then $\alpha(x) \in \mathcal{O}$ by the wing property, so α is a map from $\mathcal{O}$ to $\mathcal{O}$, though not necessarily a permutation. The *weight* of c in coordinate j is

$$w_j(c) = \sum_{x \in \mathcal{O}} [\, j \in S_x \,] + \sum_{x \in \mathcal{O}} [\, \alpha(x)_j = \pi(x)_j \,] + \sum_{t=1}^{r} d_j(\gamma_t) + 2 \sum_{e \in E_j} k_e. \tag{2}$$

The state is *admissible* if $w_j(c) = b_j$ for every j. Let $\mathcal{S}$ be the set of admissible states.

Lemma 5.2. *$\mathcal{S}$ is finite.*

Proof. All terms in (2) are nonnegative. Hence $k_e \le b_j/2$ for $e \in E_j$, and $\sum_{t=1}^{r} \sum_j d_j(\gamma_t) \le \sum_j b_j$. A permutation $\gamma \neq \mathrm{id}$ moves at least two points, and each moved point differs from its image in at least one coordinate, so $\sum_j d_j(\gamma) \ge 2$. The stack therefore has length at most $\frac{1}{2} \sum_j b_j$. Every remaining component ranges over a finite set. □

The wing involution R. Let $c = (S, \pi, \Gamma, k) \in \mathcal{S}$. Call a pair (x, j) *bad* if $j \in A_x$ and $\pi(x)_j = x_j$. If there is no bad pair, let $R(c) = c$. Otherwise let (x, j) be the first bad pair, and let $R(c)$ be c with j added to or removed from S_x.

Lemma 5.3. *R is an involution of $\mathcal{S}$. If $R(c) = c$, then π satisfies* (a) *and* (b).

Proof. The set S_x remains inside A_x because $j \in A_x$. Bad pairs depend only on π, which R does not change, so a second application undoes the first. Toggling changes only the terms of x in coordinate j in the first two sums of (2). Write $t = [\, j \in S_x \,]$. Since $\pi(x)_j = x_j$, we have $[\, \alpha(x)_j = \pi(x)_j \,] = 1 - t$, so the two terms add up to 1 before and after the toggle. Hence R preserves admissibility.

Now let $R(c) = c$. Then no pair is bad, which is (a). Fix j and x. If $j \notin A_x$, then $j \notin S_x$ and $\alpha(x)_j = x_j$, so the terms of x contribute $[\, \pi(x)_j = x_j \,]$. If $j \in A_x$, then $\pi(x)_j \neq x_j$. In that case the terms contribute 0 when $j \notin S_x$, and $1 + 1 = 2$ when $j \in S_x$, because then $\alpha(x)_j \neq x_j$ forces $\alpha(x)_j = \pi(x)_j$. Therefore

$$\sum_x [\, j \in S_x \,] + \sum_x [\, \alpha(x)_j = \pi(x)_j \,] = a_j(\pi) + 2 \sum_x [\, j \in S_x \,], \tag{3}$$

and admissibility gives $a_j(\pi) \le w_j(c) = b_j$, which is (b). □

The left involution L. Let $c = (S, \pi, \Gamma, k) \in \mathcal{S}$ and let α be as above. Note that $\alpha = \mathrm{id}$ if and only if every S_x is empty. The involution L is defined by four cases, checked in order.

(L1) *α is not injective.* Let (x, y) be the first pair with $x \neq y$ and $\alpha(x) = \alpha(y)$. Replace π by $\pi \circ (x\, y)$.

(L2) *α is a permutation and $\alpha \neq \mathrm{id}$.* Push α onto the front of the stack, replace π by $\pi \circ \alpha^{-1}$, and make every S_x empty.

(L3) *$\alpha = \mathrm{id}$ and $\Gamma = (\gamma, \Gamma')$ is nonempty.* Pop γ, put $S_x = \Delta(x, \gamma(x))$, so that the new α is γ, and replace π by $\pi \circ \gamma$.

(L4) *$\alpha = \mathrm{id}$ and Γ is empty.* Then $w_j(c) = a_j(\pi) + 2\sum_{E_j} k_e$, so $(\pi, k) \in C_0$. Replace (π, k) by $J(\pi, k)$.

Lemma 5.4. *L is an involution of $\mathcal{S}$ whose only fixed point is $c_* = (\varnothing, \pi_*, (), 0)$.*

Proof. Admissibility.

- In (L1), α is unchanged and the multiset $\{[\,\alpha(x)_j = \pi(x)_j\,], [\,\alpha(y)_j = \pi(y)_j\,]\}$ is unchanged, because $\alpha(x) = \alpha(y)$.
- In (L2), the permutation α is local and differs from id, so it is a legal stack entry. Also $\sum_x [\,j \in S_x\,] = d_j(\alpha)$, and the substitution $y = \alpha(x)$ gives $\sum_x [\,\alpha(x)_j = \pi(x)_j\,] = a_j(\pi \circ \alpha^{-1})$, which is the second sum of the new state. So the weight is redistributed without change.
- Case (L3) is the same computation read backwards. The sets $\Delta(x, \gamma(x))$ lie in A_x because γ is local.
- In (L4), J maps C_0 to itself.

Involution. Case (L1) keeps α, so the same pair is chosen again and the transposition is undone. Case (L2) produces a state with $\alpha = \mathrm{id}$ and a nonempty stack, to which (L3) applies. It pops α and restores $\pi = (\pi \circ \alpha^{-1}) \circ \alpha$ and the sets S_x. Conversely, (L3) produces a state whose $\alpha = \gamma$ is a permutation different from id, to which (L2) applies and pushes γ back. Case (L4) keeps $\alpha = \mathrm{id}$ and the stack empty, and J is an involution.

Fixed points. Moves of type (L1) change π, and moves of type (L2) or (L3) change the length of the stack, so none of them fixes a state. In (L4) the state is fixed exactly when $J(\pi, k) = (\pi, k)$, that is, when $(\pi, k) = (\pi_*, 0)$ by Proposition 4.4. The state c_* is admissible because $a_j(\pi_*) = b_j$. □

6. The construction and the proof of Theorem 2.1

Start at $c = c_* = (\varnothing, \pi_*, (), 0)$ and repeat the following steps.

(1) If no pair (x, j) with $j \in A_x$ and $\pi(x)_j = x_j$ exists, output $\tau = \pi$ and stop.

(2) Otherwise replace c by $L(R(c))$ and return to the first step.

Proof of Theorem 2.1. By Lemma 5.2 the set $\mathcal{S}$ is finite, and by Lemmas 5.3 and 5.4 the maps R and L are involutions of $\mathcal{S}$, with c_* the only fixed point of L. Lemma 3.1 shows that the construction stops at a state c with $R(c) = c$. By Lemma 5.3, its permutation $\tau = \pi$ satisfies (a) and (b). □

Remark 6.1. The proof bounds the number of iterations only by $|\mathcal{S}|$. It establishes no polynomial bound in n and $N = |\mathcal{O}|$.

7. Reimer's Butterfly Lemma

We follow the formulation of Smyth [7, Lemma 2.4], specialised to the whole cube, as recalled in Turturean's Remark 3.10 [8]. A *butterfly* is a pair $B = (a, b)$ of points with *body* a and *tip* b. Its red and yellow sets are

$$\mathrm{Red}(B) = \{b^S : S \subseteq \Delta(a, b)\}, \qquad \mathrm{Yellow}(B) = \{y : \ y_i \neq b_i \text{ for all } i \in \Delta(a, b)\}.$$

For a family $\mathcal{B}$ of butterflies, let $\mathrm{Red}(\mathcal{B})$ and $\mathrm{Yellow}(\mathcal{B})$ be the unions of the red and of the yellow sets of its members.

Corollary 7.1 (Butterfly Lemma [6])**.** *If $\mathcal{B}$ is a finite family of butterflies with pairwise distinct tips, then $|\mathcal{B}| \leq |\,\mathrm{Red}(\mathcal{B}) \cap \mathrm{Yellow}(\mathcal{B})|$.*

Proof. The argument is that of [8, Remark 3.10]. Let $\mathcal{O} = \mathrm{Red}(\mathcal{B})$. Give the tip b of each $(a, b) \in \mathcal{B}$ the wing $A_b = \Delta(a, b)$, and give every other point of $\mathcal{O}$ the empty wing. This is well defined because the tips are distinct, and A_b is a wing because $\mathrm{Red}(a, b) \subseteq \mathcal{O}$.

Let τ be the permutation of Theorem 2.1. By (a), the image $\tau(b)$ of each tip lies in $\mathcal{O} \cap \mathrm{Yellow}(a, b) \subseteq \mathrm{Red}(\mathcal{B}) \cap \mathrm{Yellow}(\mathcal{B})$. Since τ is injective, the map $(a, b) \mapsto \tau(b)$ is injective on $\mathcal{B}$. □

AI disclosure

The construction, proofs and manuscript were developed using AI models. The author scoped and directed the work and is responsible for the paper's correctness.

References

[1] S. Aaronson and A. Ambainis, *The need for structure in quantum speedups*, Theory Comput. **10** (2014), 133–166.

[2] C. Borgs, J. T. Chayes and D. Randall, *The van den Berg–Kesten–Reimer inequality: a review*, in: Perplexing Problems in Probability, Progress in Probability, vol. 44, Birkhäuser, Boston, 1999, 159–173.

[3] A. M. Garsia and S. C. Milne, *Method for constructing bijections for classical partition identities*, Proc. Natl. Acad. Sci. USA **78** (1981), no. 4, 2026–2028.

[4] L. de Moura and S. Ullrich, *The Lean 4 theorem prover and programming language*, in: Automated Deduction – CADE 28, Lecture Notes in Comput. Sci., vol. 12699, Springer, 2021, 625–635.

[5] The mathlib Community, *The Lean mathematical library*, in: Proceedings of the 9th ACM SIGPLAN International Conference on Certified Programs and Proofs (CPP 2020), ACM, 2020, 367–381.

[6] D. Reimer, *Proof of the van den Berg–Kesten conjecture*, Combin. Probab. Comput. **9** (2000), no. 1, 27–32.

[7] C. D. Smyth, *The conjectures of Rudich, Tardos, and Kusner*, Ph.D. dissertation, Rutgers University, New Brunswick, New Jersey, 2001.

[8] D. Turturean, *A proof of the Aaronson–Ambainis conjecture*, Physical Superintelligence (PSI), November 2026. `https://psi.inc/research/aaronson-ambainis/`.